

The role of parenting in children's math learning: A dual-pathway model

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Abstract

Parents' involvement in children's math learning is a critical, yet underutilized, resource. Drawing on cognitive and motivational perspectives, this chapter introduces a dual-pathway model highlighting the importance of parents' cognitive and motivational math parenting practices in fostering children's math learning. Cognitive practices include the content, level, and structure of parents' math talk and gestures. Through these practices, parents introduce numeracy concepts and scaffolded support within children's zone of proximal development, thereby providing cognitive resources for the development of children's math skills. Motivational practices include parents' autonomy-support (vs. control), positive (vs. negative) affect, and process (vs. person)-orientation around math. These practices fulfill children's basic psychological needs, which develops children's math beliefs, motivation, and engagement. Empirical evidence demonstrates that parents' cognitive and motivational practices each contribute uniquely—and in an additive fashion—to children's math learning, with the possibility of interactive effects. Proximal antecedents (e.g., attributes of children, parents, and the learning context) can shape both types of practices. By delineating distinct yet complementary mechanisms of parents' influence, the dual-pathway model offers a unified framework to inform future

research, educational practices, and family-focused interventions aimed at enhancing children's math learning.

Keywords: Learning, Math, Motivation, Parent involvement, Parenting



1. The role of parenting in children's math learning: a dual-pathway model

There is much evidence that parents play a significant role in children's learning (for a recent review, see [Grolnick & Pomerantz, 2022](#)). In fact, parents have been deemed an underutilized resource in fostering children's math learning (e.g., [Harackiewicz et al., 2012](#)), which is declining in the United States ([National Science Board & National Science Foundation, 2024](#)). Much of the theory and research on the role of parenting in children's math learning has focused on the idea that parents can advance children's development of math skills prior to formal schooling by providing cognitive input (e.g., [Gunderson & Levine, 2011](#)). In this work, parents' provision of cognitive resources (e.g., numerical concepts) is the central pathway by which parents can support children's math learning, with much evidence that indeed parents' cognitive practices (e.g., relatively advanced math talk) matter (e.g., [Eason et al. 2021](#); [Ramani et al., 2015](#)). A large body of theory and research, however, indicates that parents can provide children with motivational resources (e.g., feelings of autonomy) that foster their learning (for a review, see [Pomerantz, Kim, & Cheung, 2012](#)). Significantly, parents' motivational practices (e.g., supporting children's autonomy in solving math problems) have recently been identified as contributing to children's math learning (e.g., [Oh et al., 2022](#); [Wu et al., 2022](#)). Thus, the cognitive pathway may not be the sole pathway by which parents shape children's math learning ([Wu et al., 2024](#)).

In this chapter, we introduce a dual-pathway model of the role of parenting in children's math learning by integrating the theory and research on parents' practices assumed to afford cognitive resources with that on their practices assumed to afford motivational resources. In brief, as shown in [Fig. 1](#), although a central pathway by which parents can contribute to children's math learning is via their cognitive practices, they can also contribute via their motivational practices. The cognitive and motivational pathways provide distinct resources to children that can together optimize children's math learning over the course of their development. In line with ecological theories of development in which children, parents, and

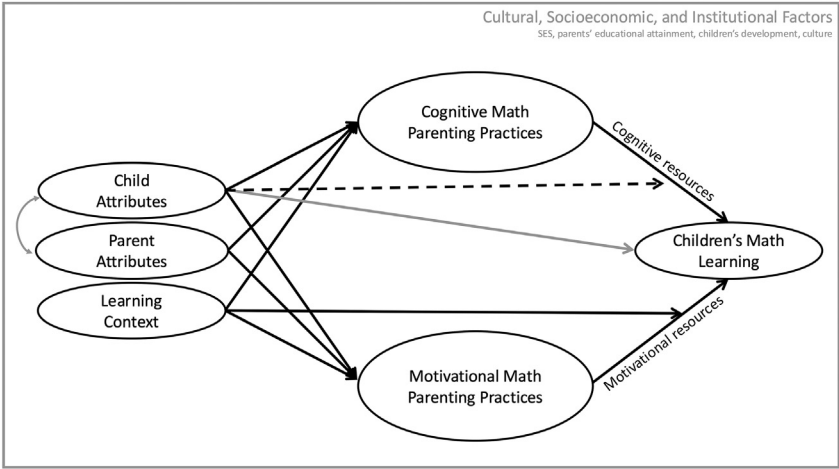


Fig. 1 The dual-pathway model of the role of parenting in children’s math learning. *Note.* The dashed arrow indicates that there is no empirical evidence for the link. The gray arrows, lines, and text reflect processes that are present in the socialization process but not discussed here due to significance or a lack of theoretical and empirical work.

the multiple social environments in which they reside contribute to the socialization process (e.g., [Belsky, 1984](#); [Brofenbrenner & Morris, 2006](#)), the model includes forces at multiple levels that may shape parents’ cognitive and motivational practices and thus children’s math learning. In delineating the model, we begin by discussing the cognitive and motivational resource parenting pathways. We then turn to the most proximal forces (i.e., attributes of children, parents, and the learning context) that may shape the two pathways. We end with a discussion of key directions for future research and implications for educators.

➤ 2. Cognitive and motivational resource parenting pathways

The central idea behind the dual-pathways model is that both parents’ cognitive and motivational practices contribute to children’s math learning by providing distinct resources to children that build distinct strengths. In brief, parents’ cognitive practices in the math context (for types and definitions, see [Table 1](#)) convey or support engagement with math concepts among children. Consequently, such practices may directly develop children’s math skills. Parents’ motivational practices in the math context (for types

Table 1 Cognitive math parenting practices: types, definitions, and evidence.

Type	Definition	Evidence Summary
Math talk	Verbal references to math concepts, such as counting, number recognition, and problem-solving.	There is a small positive association between parents' math talk and children's math learning (for reviews, see Daucourt et al., 2021 ; Silver et al., 2024), but findings are limited and vary across studies (e.g., Levine et al., 2010 ; Ren et al., 2022 ; Thippana et al., 2020).
Level of math talk	The complexity of the math content or strategies discussed with children, ranging from basic concepts to more advanced concepts relative to what children are learning.	Advanced math talk, within children's range of learning, predicts enhanced math abilities and engagement (e.g., Casey et al., 2018 ; Gunderson & Levine, 2011 ; Ramani et al., 2015 ; Wu et al., 2024).
Structure of math talk	Prompts (i.e., questions, directions, and hints that encourage math engagement and problem-solving) or statements (i.e., providing information, facts, and instructions related to math concepts) about math.	Parents' prompts, particularly advanced prompts, predicts more advanced math talk, engagement, and skills among children. Parents' statements have less positive, and even negative, effects (e.g., Eason et al., 2021 ; Kurkul & Corriveau, 2018 ; Wu, Oh, Hyde, & Pomerantz, 2024)
Math gestures	Use of hand movements to illustrate or represent math concepts during interactions with children.	Gestures used in combination with math talk is associated with children's enhanced math engagement with key math concepts (i.e., cardinality principle, counting strategy; e.g., Congdon et al., 2017 ; Goldin-Meadow et al., 2012); (Gibson, Gunderson, & Levine, 2020)).

and definitions, see [Table 2](#)) allow children to meet basic psychological needs, such as autonomy and relatedness, which develop children's math beliefs, motivation, and engagement. Importantly both sets of strengths—that is math skills and math beliefs, motivation, and engagement—appear to contribute to children's math achievement, at least by the end of the

Table 2 Motivational math parenting practices: types, definitions, and evidence.

Type	Definition	Evidence summary
Autonomy support (vs. control)	Autonomy support includes offering choices, encouraging independent problem-solving, and providing guidance only when needed. Control entails pressure, excessive direct instruction, problem-solving on behalf of children, or coercive tactics to ensure compliance.	Parents’ autonomy support (vs. control) predicts higher math achievement in children (e.g., Cimon-Paquet et al., 2020 ; Oh et al., 2022 ; Silinskas & Kikas, 2019a).
Affect	Emotional tone ranging from positive (e.g., happiness, joy, and love) to negative (e.g., irritation, anxiety, and sadness).	Parents’ positive (vs. negative) affect predicts heightened math motivation, engagement, and achievement (e.g., Silinskas & Kikas, 2019b ; Wu, Barger, Oh, & Pomerantz, 2022).
Responses to children’s performance	Process-oriented responses focus on emphasizing internal, controllable factors such as effort and strategy use. Person-oriented responses highlight innate ability or intelligence, implying that performance reflects stable, inherent traits.	Process-oriented responses predict improved math achievement (e.g., Gunderson et al., 2018). Person-oriented responses predict children’s heightened math anxiety, dampened preference for math challenge, and lower math achievement (e.g., Barger et al., 2022).

elementary school years and continuing into middle and high school years (e.g., [Jordan et al., 2009](#)).

2.1 Cognitive parenting practices in the math context

The sociocultural theory, originally proposed by [Vygotsky \(1978\)](#), posits that social interactions with more knowledgeable others, such as parents, play a critical role in children’s cognitive development ([Gauvain, 2001](#); [Rogoff, 1990](#)). Parents’ use of language has been emphasized in these interactions as the “cultural tools” that help children make meaning and develop abstract concepts ([Vygotsky & Cole, 1978](#), p. 123). As such, parents’ math talk (i.e., verbal reference to a variety of concepts involved in math) features prominently in the sociocultural perspective. It has been argued that through such talk parents provide information about key math concepts that

facilitate children's math learning (e.g., [DePascale et al. 2021](#)). In addition, borrowing from the bootstrapping account in the language acquisition area (e.g., [MacWhinney, 2014](#)), [Gunderson & Levine \(2011\)](#) suggest that parents' cognitive input, often via their math talk, matters for the development of children's math skills (e.g., numerical knowledge and vocabulary). This aligns with broader cognitive theories of learning, which postulate that more frequent exposure to numerical concepts helps children build mental representations that gradually refine their math understanding over time (e.g., [Sweller, 1988](#)).

Despite the theoretical importance of parents' math talk, research on its impact has been mixed. Some studies support the notion that frequent exposure to early math talk enhances children's math skills by the time they enter school (e.g., [Levine et al., 2010](#); [Susperreguy & Davis-Kean, 2016](#)), whereas others do not (e.g., [Ren et al., 2022](#); [Thippana et al., 2020](#)). Evidence from meta-analyses also reveals limited support for this association. [Daucourt et al. \(2021\)](#) meta-analysis of nine studies with children prior to entering formal schooling reported a near-zero association ($r = 0.03$) between parents' math talk and children's math achievement. A more recent and comprehensive meta-analysis ([Silver et al., 2024](#)) including 22 studies of the effects of parents' math talk with children from two to seven years old, found a small, but statistically significant, effect ($b = 0.10$, $SE = 0.03$, $p = 0.002$). When children's general cognitive abilities or prior math performance were taken into account, however, this effect was reduced in size becoming non-significant. These findings suggest that simply engaging in math talk frequently may not be enough.

Indeed, consistent with [Vygotsky's \(1978, 1986\)](#) ideas on scaffolding, the content of parents' math talk matters. More advanced content within the range of children's developing skills pushes children toward what [Vygotsky \(1986\)](#) called the "zone of proximal development" (p. 86). The cognitive perspective also suggests that exposure to relatively advanced concepts can motivate children to understand more cognitively demanding numeracy concepts (e.g., [Gunderson & Levine, 2011](#)). In line with this view, it appears that parents' more advanced numeracy talk, within the range of children's developing skills, benefits children's math learning (e.g., [Casey et al., 2018](#); [Ramani et al., 2015](#)). For example, [Gunderson and Levine \(2011\)](#) found that during observations of natural parent-child interactions when children were one to two-and-a-half years old, the amount of numeracy talk parents devoted to sets (i.e., counting or labeling sets of present objects) consisting of higher (i.e., from 4 to 10), but not lower (i.e.

from 1 to 3), numbers of items, was associated with four-year-old children's cardinal-number knowledge, even after adjusting for parents' socioeconomic status. Similarly, parents' focus on more advanced numerical concepts—such as cardinality, ordinal relations, and arithmetic—is associated with stronger numerical reasoning and magnitude comparison skills among preschool children, over and above parents' other teaching practices (Ramani et al., 2015).

Although the level of parents' math talk is important, Vygotsky's (1978, 1986) ideas on scaffolding suggest the structure of such talk also matters. In this regard, recent research guided by the sociocultural perspective has examined parents' prompts (i.e., questions, directions, and hints, often to encourage children's engagement) and statements (e.g., provision of information, facts, and instructions, often to convey relevant concepts) in the context of their math talk with children (e.g., Eason et al. 2021; Wu et al., 2024). Drawing from Vygotsky (1986) and Rogoff (1998), Eason et al. (2021) make the case that prompts may be more likely than statements to push children into the zone of proximal development by providing the opportunity for children's more advanced engagement. Parents' prompts may nudge children toward active engagement in the activity at hand as they often require a response that relies on exploration and operations (e.g., Vandermaas-Peeler et al., 2016). Parents' statements, in contrast, may provide useful input on numerical concepts, but children's reception may be relatively passive with minimal exploration and operations on the concepts (Wu et al., 2024). Similarly, Sigel et al., (2002) "distancing strategies" framework highlights that cognitively demanding prompts—such as those requiring children to predict, deduce, or reason through problems—are particularly effective in advancing learning.

The empirical evidence is consistent with these ideas in that it shows that parents' prompts, but not necessarily their statements, are linked to advanced math engagement. For example, studying mothers and their two- to four-year-olds, Eason et al., (2021) found that during activities such as pretend cooking, brick building, and math story reading, the more parents used both prompts and statements, the more children engaged in math talk, but only prompts were linked to greater diversity in children's numerical vocabulary. A similar pattern has been observed among parents and their children in early elementary school. Using a minute-to-minute within-dyad analytic approach, Wu et al., (2024) showed that when both the level and structure of parents' math talk (i.e., prompts and statements) were examined simultaneously, parents' advanced prompts, but not statements, were associated with children's subsequent engagement in math strategies,

which were often advanced as well, even after taking into account children's engagement the prior minute.

Parents' provision of cognitive resources often extends beyond verbal input to include non-verbal elements such as gestures. From a cognitive perspective, gestures play a fundamental role in shaping children's math understanding by providing an additional modality for processing math concepts (e.g., [Butterworth, 1999](#); [Fuson, 1988](#)). For example, [Fuson \(1988\)](#) makes the case that gestures can function as a cognitive bridge between verbal number knowledge and abstract numerical representations, reinforcing key counting principles such as one-to-one correspondence and stable order. Pointing gestures, for instance, can help children track number words during counting, ensuring accurate item-to-number mapping and supporting early math development (e.g., [Wiese, 2003](#)). In parent-child interactions around math, gestures frequently accompany math talk and act as a supplementary channel of cognitive input, enhancing the communication of both foundational and advanced math concepts (e.g., [Flevaris & Perry, 2001](#); [Goldin-Meadow, 2009](#)). Although research on parents' gestures is still emerging, early findings suggest they provide symbolic cognitive support, improving eight- to nine-year-old children's comprehension of math concepts (e.g., [Congdon et al., 2017](#); [Wakefield et al., 2019](#)).

2.2 Motivational parenting practices in the math context

Parents' involvement in children's math learning can also confer benefits on children by providing them with motivational resources that foster their math learning (e.g., [Oh et al., 2022](#); [Wu et al., 2022, 2024](#)). Self Determination theory (SDT; [Deci & Ryan, 1985; 2000](#)) posits that people have three fundamental psychological needs from an early age: autonomy, competence, and relatedness. Parents' involvement in children's learning process is an important context for meeting these needs and can provide children with the motivational resources that facilitate math learning (e.g., [Grolnick et al., 1991](#); [Grolnick & Pomerantz, 2022](#)). Parents' involvement is especially beneficial when parents adopt practices that are autonomy supportive (vs. controlling), affectively positive (vs. negative), and process (vs. person) oriented (for a review, see [Pomerantz, Kim, & Cheung, 2012](#)).

Parents' involvement can be characterized as autonomy supportive, in that parents provide children with choices, allow them to take initiative, and adopt children's perspectives. The more parents' involvement is autonomy supportive, the more children feel agentic and competent, feelings that

ultimately contribute to their intrinsic motivation (e.g., [Lerner et al., 2022](#); [Ryan & Deci, 2000](#)). In the math context, parents can be autonomy supportive by letting children lead the way when solving problems and only providing hints when children ask for help, allowing children to feel autonomous and capable of doing math on their own. Parents' involvement can also be characterized as controlling in that parents pressure, direct, and restrict children (e.g., [Grolnick & Pomerantz, 2022](#); [Grolnick et al., 1991](#)). Controlling parenting has been argued to undermine children's academic adjustment, as it dampens children's feelings of autonomy and competence (e.g., [Grolnick et al., 1991](#); [Grolnick & Pomerantz, 2009](#)). Parents' control in the math context may manifest as directing children on how to solve problems or intervening early in the problem-solving process, which may lead children to feel helpless and incapable when it comes to math, thereby fostering their disengagement from math.

Although there is much evidence that parents' autonomy support (vs. control), contributes to children's motivation, engagement, and learning in general (e.g., [Cheung et al., 2016](#); [Grolnick et al., 2002](#)), only recently has research examined the effects of such parenting in the math domain specifically. The evidence to date indicates that the more parents are autonomy supportive and the less they are controlling in children's math learning, the higher elementary and middle school children's perceptions of their math competence, persistence with math homework, and math achievement (e.g., [Silinskas & Kikas, 2019a](#); [Cimon-Paquet et al., 2020](#)). For example, [Wu et al., \(2024\)](#) found that the more autonomy supportive (vs. controlling) parents were in working on math activity with their early elementary school children one minute predicted enhanced engagement in the activity among children the following minute adjusting for their engagement the prior minute. This may be especially true for children struggling with math as [Oh et al., \(2022\)](#) found that parents' autonomy support predicted increases over time in children's math achievement largely among children who were initially low achieving in math.

Parenting is thought to be imbued with affect ([Dix, 1991](#)), as parents' involvement in the learning process (e.g., helping with homework) can be enjoyable as well as frustrating for both children and parents (e.g., [Pomerantz et al., 2005a](#); [Wu et al., 2022](#)). On the one hand, parents' involvement can be affectively positive, in that parents display positive emotions such as happiness and joy. This affect may help children experience and view math learning positively, which may foster beliefs, motivation, and engagement that support math achievement (e.g., [Else-Quest et al., 2008](#);

Pomerantz et al., 2005a). On the other hand, parents' involvement can be affectively negative, as characterized by their display of negative emotions such as irritation and frustration, which may communicate to children that math is unpleasant and should be avoided, potentially diminishing children's beliefs, motivation, and engagement (e.g., Nolen-Hoeksema et al., 1995; Pomerantz et al., 2005a).

When parents' involvement is more affectively negative (vs. positive), elementary school children are more likely to exhibit helplessness and a weaker inclination to take on challenges (e.g., Estrada et al., 1987; Nolen-Hoeksema et al., 1995). For example, mothers' negative affect when assisting with eight- to 12-year-old children's homework predicts children's dampened motivation six months later; this effect, however, was reduced when mothers also reported greater positive affect when assisting children with homework (Pomerantz et al., 2005a). Research in the math domain is limited, with evidence that parents' negative, but not positive, affect contributes to children's math learning. Namely, Wu et al., (2022) found that parents' negative affect when helping first and second graders with math homework predicted children's dampened math liking, preference for math challenge, and achievement one year later, above and beyond children's prior math adjustment and parents' educational attainment.

Parents' responses to children's performance also can play a role in children's math learning. Some parents focus their responses on the process of learning, in that they emphasize effort and strategy, whereas others rely on person-oriented responses that focus on innate attributes of children (e.g., their math ability; for reviews, see Henderlong & Lepper, 2002; Pomerantz, Kim, & Cheung, 2012). Process-oriented responses encourage children to persevere through challenges and build their skills, thereby enhancing children's motivation and engagement. In contrast, person-oriented responses can have the opposite effect on children's learning. Given that math is often viewed as requiring innate talent (Leslie et al., 2015), parents may view children as either having math ability or not, leading them to adopt more person-oriented responses to their children's success and failures in math. These responses may signal to children that their performance reflects their ability, so that children come to equate low performance with low competence, undermining their motivation and engagement (e.g., Mueller & Dweck, 1998).

In line with this perspective, research suggests that parents' process-responses facilitate children's math learning, whereas their person-responses undermine it (e.g., Gunderson et al., 2013). Gunderson et al., (2018) found that parents' process-oriented, but not person-oriented, responses to toddlers'

successes during everyday activities predicted stronger entity theories and greater preference for challenge in second and third grade, which predicted higher math achievement one year later, controlling for children's prior achievement. Among older children, parents' person-oriented, but not process-oriented, praise of children's day-to-day academic successes predicts 10-year-old children's stronger entity theories of academic ability and lower preference for challenge in school six months later, accounting for children's earlier theories and preferences (e.g., [Pomerantz & Kempner, 2013](#)). Similarly, parents' person, but not process, responses to first and second graders' math successes and failures predict children's heightened math anxiety, dampened preference for math challenge, and lower math achievement one year later, adjusting for a variety of potential confounds ([Barger et al., 2022](#)). Variation in the effects of process- and person-oriented praise may stem from differences in the contexts in which these responses were recorded.

Taken together, these findings highlight the role of parents' motivational practices in supporting children's math learning. However, drawing from Parent $\times$ Child socialization models (e.g., [Bates et al., 1998](#); [Kochanska, 1993](#)), [Pomerantz et al. \(2005b\)](#) have made the case that children who have a heightened need for motivational resources may be particularly sensitive to their parents' motivational practices such that they are particularly likely to benefit from practices such as autonomy support that afford such resources and particularly likely to suffer from practices that do not. In line with this idea, elementary school children who struggle in school and thus may have relatively depleted motivational resources are particularly at risk in the face of controlling or affectively negative parenting (e.g., [Pomerantz et al., 2005a](#); [Ng et al., 2004](#)). In the math domain, [Oh et al. \(2022\)](#) found that the more controlling parents were when interacting with their six- and seven-year old children around math, the poorer children's math achievement one year later but only among children who were already struggling with math.

2.3 Additive and interactive effects of cognitive and motivational parenting practices in the math context

The dual-pathway model we present here portrays parents' cognitive and motivational practices and the pathways by which they influence children's math learning as generally distinct with independent effects. It is possible, however, that the two types of practices could overlap and thus not have independent effects. This is most likely in regard to the structure

of parents' math talk (i.e., prompts and statements) and parents' autonomy support (vs. control). For example, operationalizations of autonomy-supportive parenting often include prompts (e.g., [Grolnick et al., 2002](#); [Oh et al., 2022](#)) as they can be instrumental in scaffolding children's exploration and treating children as experts. Prompts, however, are only one component of autonomy-supportive parenting and only sometimes reflect such parenting—for example, when they are used in a manner providing children with agency. Similarly, operationalizations of controlling parenting often include statements (e.g., [Grolnick et al., 2002](#); [Oh et al., 2022](#)) given that statements can be directive; but statements are also only one component of parents' control, and when they are part of responsiveness and support (e.g., used in response to children's request for assistance), are not controlling. As [Wu et al. \(2024\)](#) point out, autonomy support and control are also broader (e.g., they include parents' general child- vs. parent-centered practices) than prompts and statements, which in the context of math talk refer specifically to math concepts.

In line with the idea that parents' cognitive and motivational practices are generally distinct, [Wu et al. \(2024\)](#) found only modest associations between prompts and autonomy support ($r = 0.23, p < 0.001$) and control ($r = -0.18, ps < 0.001$) and statements and autonomy support ($r = 0.00, ns$) and control ($r = 0.24, p < 0.001$) among parents of early elementary school children during a math activity. Importantly, these cognitive and motivational practices also made distinct contributions to children's engagement during the activity. For example, suggestive of the transmission of cognitive resources (e.g., advanced math concepts), parents' advanced prompts one minute predicted children's engagement, particularly of advanced math strategies, the next minute over and above not only children's engagement the prior minute, as described above, but also parents' autonomy support and control. In contrast, parents' autonomy support (vs. control) appeared to provide motivational resources that set the foundation for sustained persistence as such parenting foreshadowed children's engagement, but not necessarily of advanced strategies, the next minute adjusting for children's engagement the prior minute as well as parents' prompts and statements. Other overlaps among parents' cognitive and motivational practices have not been examined but are possible—for example, prompts may be more likely in the context of process praise as parents encourage children to think through their effort and strategies.

Although the evidence to date suggests parents' cognitive and motivational practices are likely to provide distinct resources and thus have additive

effects, it is also possible the two types of practices and pathways have interactive effects (i.e., one modulates the other's contribution). For example, when parents combine advanced statements with autonomy support, they may provide useful cognitive resources that children can experiment with and apply with agency. It could also be that children are more open to parents' statements when parents establish feelings of relatedness via their positive affect. The one study that has examined the interactive effects of the structure and level of parents' math talk with their autonomy support (vs. control), however, did not find evidence for interactive effects for early elementary school children (Wu et al., 2024). Similarly, Ren et al. (2022) did not find interactive effects of parents' spatial talk and the frequency of their praise for early elementary school children.



3. Antecedents of cognitive and motivational resource parenting pathways

Given that both parents' cognitive and motivational practices can be important to children's math learning, a key issue is identifying what undergirds the two so as to understand how schools can best support parenting that facilitates children's learning. Although ecological theories of development (e.g., Belsky, 1984; Brofenbrenner & Morris, 2006) suggest parents' practices may be shaped by a set of complex multi-level forces, we focus here on the three most proximal forces—attributes of children, parents, and the math learning context—as there has been the most theoretical and empirical attention to these forces. Moreover, such forces may be particularly easy for educators to target to facilitate the most constructive parenting.

3.1 Children's attributes

Although much of the theory and research on parents' involvement in children's learning has assumed a model in which parents set the course of children's learning via the resources they provide (e.g., Gunderson & Levine, 2011; Grolnick & Pomerantz, 2022), children are rarely passive recipients in the socialization process (e.g., Collins et al., 2000; Sameroff, 2009). Instead, they can play an active role in shaping their own learning through interactions with their environment (e.g., Oh et al., 2022; Pomerantz et al., 2005b). Piaget (1976), in his theory of cognitive development, similarly emphasized that children construct knowledge by engaging with the world around them. One of the most significant ways children influence their own

learning is via the attributes they bring to interactions with parents as these attributes can shape the ways in which parents are involved in children's learning (e.g., [Belsky, 1984](#); [Pomerantz et al., 2005b](#)).

Although a variety of children's attributes (e.g., temperament) have been deemed important drivers of the socialization process (e.g., [Kiff et al., 2011](#); [Sanson et al., 2004](#)), in the learning context, children's competence-related attributes (i.e., capacity to understand, interpret, and engage with the academic subject at hand) are key given that parents' involvement in children's learning often revolves around such attributes—for example, children's capacities may determine the extent to which they can work independently (e.g., [Pomerantz et al., 2005b](#)). Children's competence-related attributes in the math context may be important drivers of both parents' cognitive and motivational practices but may operate differently for the two. The more children lack the capacity for math, the more parents may adjust their cognitive practices so that they provide children with the appropriate cognitive resources—an essential part of [Vygotsky's \(1986\)](#) concept of scaffolding. At the same time, however, parents' motivational practices may become less constructive such that they provide fewer motivational resources to children who lack capacity and thus may be most in need of such resources (e.g., [Moorman & Pomerantz, 2008](#); [Pomerantz et al., 2005b](#)).

Cognitive parenting practices. [Vygotsky \(1986\)](#) highlights how parents scaffold children's learning by providing support that is appropriately challenging for children's current skill-level. In practice, children's math competence serves as a cue for parents to adjust their cognitive practices, tailoring their support to match children's developing needs (e.g., [DePascale et al., 2021](#); [Susperreguy et al., 2020](#)). Parents provide cognitive input that aligns with their child's developmental stage, gradually increasing the complexity of the math concepts they introduce. For example, research on children aged two to four years, who are in the early stages of math learning, shows that parents primarily focus on foundational skills, such as counting and number recognition, with minimal introduction of more complex concepts like magnitude comparison or calculations ([Vandermas-Peeler et al., 2012](#)). However, as children grow older and become proficient counters (around ages six to seven years), parents appear to naturally reduce their emphasis on these basic skills and shift toward more advanced numerical discussions (e.g., [Wu et al., 2024](#); [Zippert et al., 2019](#)).

Beyond developmental trends, parents also respond dynamically to their individual children's competence, adjusting their cognitive input in real-time. Although direct evidence on this process remains limited, some studies

have found concurrent associations in preschool-aged children, suggesting that the more competent children are, the more parents provide more advanced math input, while referencing foundational concepts for those who are less competent (e.g., [DePascale et al., 2021](#); [Vandermaas-Peeler et al., 2012](#)). Two recent studies are suggestive of this pattern. Using vignettes in which children provided correct and incorrect responses to math problems, [Çarkoğlu and Eason \(2025\)](#) found that parents said they would provide more elaborative cognitive support—such as explanations and follow-up questions—after errors than after correct responses. [Wu et al. \(2025\)](#) applied a minute-to-minute within-dyad analytic approach to examine how first- and second-grade children's correct and incorrect use of strategies predicted parents' math talk from one minute to the next, while accounting for parents' prior talk. Parents provided more elaborative cognitive input when children made mistakes, offering both basic statements and prompts to support children's skill development. When children used strategies correctly, however, parents increased their use of advanced prompts to encourage higher-level math thinking.

Motivational parenting practices. Although parents may adjust their cognitive practices to children's capacity such that they provide children with optimal cognitive resources, this is not the case when it comes to parents' motivational practices. When children are having difficulty, parents may be most likely to use motivational practices with children that detract from the children's already sparse motivational resources. When children struggle with math, parents may feel pressure to intervene to ensure children meet performance standards, which can affect their motivational practices (e.g., [Pomerantz et al., 2005b](#)). Drawing on the SDT framework, [Grolnick \(2003\)](#) argues that “pressure from below”, which can manifest in a lack of capacity among children, can undermine parents' motivational practices such that they are, for example, less autonomy supportive (vs. controlling) and affectively positive (vs. negative) in their interactions with children around learning.

Consistent with this idea, several studies find that when children struggle academically, parents respond by increasing their control (vs. autonomy support) and negative (vs. positive) affect. For example, in a minute-to-minute observational study conducted by [Moorman and Pomerantz \(2008\)](#), when four-year-olds displayed helpless (vs. mastery) behavior during a challenging cognitive task, mothers responded with heightened control the following minute over and above their practices the prior minute. Experimental research further supports this pattern, showing that parents

become more controlling following fifth and sixth grade children's induced failure (vs. success) (Wuyts et al., 2017). Within the domain of math learning, longitudinal research finds that the lower children's math achievement in third grade the greater the increase in parents' controlling homework assistance by sixth grade (Silinskas & Kikas, 2019b). In addition, Silinskas et al., (2015a) found that children's lower math achievement in first grade predicted increased negative affect and decreased positive affect among parents when assisting with children's math homework between second and fourth grade. Unfortunately, to our knowledge, the role of children in parents' process (vs. person) praise has not been examined.

3.2 Parents' attributes

Parents join children in bringing their own attributes to their interactions with children in the learning context. Parents' cognitions such as their attitudes, beliefs, goals, and expectations have been identified as contributing to a variety of parenting practices (for reviews, see Bornstein, 2015; Darling and Steinberg, 1993; Darling and Steinberg, 2017). When it comes to learning, parents' competence-related cognitions have been considered as particularly important (e.g., Carkoglu et al., 2023; Eccles et al., 1983). Parents' competence-related cognitions can be focused on children's competence (e.g., as is the case for parents' perceptions of children's competence) as well as on their own competence (e.g., as is the case for their anxiety about their competence).

Particularly central in the theory and research on parents' socialization of children's learning has been Eccles' (1983) Expectancy Value model which posits that parents' perceptions of children's competence play a key role in shaping children's learning by acting as self-fulfilling prophecies with parents' treating children in line with their perceptions, regardless of their accuracy. Indeed, a large body of evidence indicates that the more parents view their children as possessing competence, including in the math domain, the more positive children's perceptions of competence and the higher their achievement over time, over and above children's prior perceptions and achievement (e.g., Simpkins et al., 2012; for a review, see Pinquart & Ebeling, 2020). This may be due in part to the influence parents' perceptions of children's competence have on their cognitive and motivational practices in the learning context (e.g., Skwarchuk et al., 2014; Susperreguy et al., 2020).

Beyond parents' perceptions of children's competence, parents' beliefs about their own competence as manifest in their math anxiety (i.e., fear or tension around math) which also includes an affective component, have

gained increasing attention for their role in shaping children's math learning. Mounting evidence suggests that parents' math anxiety is predictive of children's heightened math anxiety and dampened math achievement (e.g., Maloney et al., 2015; Szczygiel, 2020), even after controlling for children's prior math adjustment (e.g., Oh et al., 2022). This is likely because math anxious parents may engage less frequently in math-related activities, and when they do, they may struggle to provide the cognitive and motivational resources necessary for fostering children's math learning (for a review, see Carkoglu et al., 2023).

Cognitive parenting practices. Because parents' perceptions of children's competence are often based, at least partially in reality, they may act much like children's actual competence. Indeed, parents' perceptions may serve as a cue for parents to adjust their cognitive practices, tailoring their support to match children's developing needs. A growing body of research examines how parents' perceptions of children's math competence are associated with their cognitive practices (for a review, see Douglas et al., 2021). The more positively parents perceive children's math competence and the more positive their expectations, the more frequent and advanced their math interactions with children (e.g., Skwarchuk et al., 2014; Susperreguy et al., 2020; Zippert & Ramani, 2017). For example, Uscianowski et al. (2020) found that parents with more positive perceptions of preschool children's math competence were more likely to indicate that they would use more complex math talk when reading a math book to children.

Parents' math anxiety may also play a role in their cognitive parenting practices in the math context. Individuals with math anxiety often avoid engaging in math-related activities in an effort to preserve their feelings of self-worth and control (e.g., Ashcraft & Moore, 2009; Hembree, 1990). Thus, parents with heightened math anxiety may engage in fewer math-related interactions, such as math talk, thereby limiting opportunities for children to develop essential math skills. Indeed, parents' heightened math anxiety is associated with less frequent math engagement and reduced math talk (e.g., Berkowitz et al., 2021; Kiss and Vukovic, 2021). For example, del Río et al. (2017) found that parents' heightened math anxiety was negatively associated with the frequency of their engagement with advanced home numeracy practices with kindergarten children, with such practices associated with lower math achievement among children. Similarly, parents' math anxiety has been associated with less frequent cardinal number talk with children under the age of three, particularly among higher socioeconomic families (Berkowitz et al., 2021).

Motivational parenting practices. It is also possible that parents' perceptions of their children's math competence may contribute to motivational parenting practices. However, despite theoretical support, research on the role of parents' perceptions of children's competence in shaping motivational parenting practices is practically nonexistent. In contrast, the role of parents' math anxiety in parents' motivational practices in the math learning context has received a fair amount of attention. Math-anxious parents may often feel incapable when working with children on math, even when it involves only rudimentary problems such as adding one- or two-digit numbers. Math-anxious parents are prone to frustration and stress when assisting children with math (DiStefano et al., 2020), which may lead to heightened negative (vs. positive) affect and control (vs. autonomy support). To avoid having to be involved in math, math-anxious parents may also be controlling to ensure children finish quickly (Oh et al., 2022) and because they are inflexible when it comes to math (Maloney et al., 2015; Retanal et al., 2021).

In line with these ideas, a concurrent study found that the more parents of sixth and eighth grade children were math anxious, the more they reported their assistance with children's math homework as being controlling and affectively negative (Retanal et al., 2021). Studying first and second grade children, Oh et al., (2022) found that math anxious parents were less autonomy supportive and were more controlling in assisting with math—both in the homework context and in a math game, particularly when children struggled with math, presumably because assisting children who struggle with math can be particularly challenging for parents who themselves feel anxious about their math competence and would prefer to avoid math altogether.

3.3 Learning context attributes

Parents' involvement can also be shaped by the attributes of the contexts in which learning takes place. One of the most widely studied context attributes in the math domain is the distinction between formal and informal learning contexts. Research has long categorized home numeracy activities along this dimension, drawing from similar classifications in the literacy domain (e.g., Skwarchuk et al., 2014). According to Skwarchuk et al., (2014), formal math activities involve structured, goal-directed interactions in which parents deliberately teach children numerical concepts, such as numbers, quantity, or arithmetic, with the explicit purpose of enhancing numeracy skills. These activities often take the form of completing math homework, worksheets, or flashcards.

In contrast, informal math activities occur in everyday experiences where math-related learning happens incidentally rather than as a primary objective. These activities include such things as playing number board games, engaging in measurement tasks during cooking or crafting, making quantity comparisons, and spatial reasoning games. In these contexts, parents may introduce math concepts more spontaneously and in ways that are naturally embedded in the activity at hand (e.g., [Elliott & Bachman, 2018](#); [Skwarchuk et al., 2014](#)). These differences are not trivial as they have implications for both parents' cognitive and motivational practices.

Cognitive parenting practices. The math learning context plays a role in shaping the cognitive resources parents provide, for example, via the structure and level of their math talk (e.g., [Bjorklund et al., 2004](#); [Eason & Ramani, 2020](#); [Zippert et al., 2019](#)). In structured, instruction-focused formal activities such as homework, parents have clear opportunities to observe children's math skills in action and provide targeted scaffolding. The structured nature of these activities may guide parents in aligning their cognitive input with the specific math content their child is learning. Moreover, instructional materials for these activities, such as worksheets, can serve as a reference, helping parents provide developmentally appropriate support through math prompts and statements.

In contrast, because math is not the explicit focus of informal activities, parents may have fewer opportunities to engage in structured math talk leading their involvement to be more spontaneous and context-dependent. Even when math-related moments arise, parents may lack the necessary guidance or control to tailor their cognitive support to the level of the material or children's skills. As a result, informal settings may lead parents' math talk to be less instructional, with the level of the concepts introduced typically depending on the activity itself, often centering on basic numerical concepts, rather than the more advanced topics children encounter in school (e.g., [Çarkoğlu & Eason, 2025](#); [Zippert et al., 2019](#)).

Research is in line with this perspective. For example, [Bjorklund et al., \(2004\)](#) found that when engaging five-year-old children in structured formal math learning contexts, parents provided significantly more prompts and cognitive guidance compared to when they played informal board games with children. Similarly, [Eason and Ramani \(2020\)](#) found that parents of four- and five-year-olds engaged in more math talk in structured, guided learning contexts than in unguided contexts, even when provided with the same materials. Among this same group of children, [Çarkoğlu and Eason \(2025\)](#) presented vignettes to parents in which children provided

incorrect responses to math problems. They found that parents were more likely to correct errors directly in formal learning contexts, reinforcing accurate numerical understanding, whereas in informal settings, they often ignored or reframed mistakes in a playful manner. These findings suggest that structured learning environments may naturally elicit more cognitively supportive practices from parents.

Beyond the formal-informal distinction, other context attributes can also shape parents' cognitive involvement. Just as structured tasks like homework can guide parents in delivering targeted cognitive input, other math-rich environments can similarly support parents in integrating math talk into their interactions with children. Specifically, the nature of the materials used in math activities can influence how parents and children interact. For example, [Masek et al. \(2024\)](#) found that certain materials, such as grocery shopping sets and picture books, encouraged richer math talk among parents and their 24- to 36-month-old children compared to manipulatives like shape sorters or magnet board games. The richness of math talk in these contexts was positively associated with children's understanding of number words. Additionally, [Zippert et al. \(2019\)](#) found that when parents were given additional guidance on how to interact with a numerical tablet game, they engaged in significantly more math-related talk with their children.

Motivational parenting practices. Although formal learning contexts can provide rich opportunities for parents to offer cognitive input, they may not necessarily foster constructive motivational practices given the pressure parents may experience in formal contexts which are often evaluative. SDT ([Deci & Ryan, 1985](#)) suggests that contexts that emphasize evaluation and performance over learning can lead parents to feel pressured, ultimately shaping how they get involved with their children. Among formal learning activities, math homework stands out as a unique and highly structured context that becomes increasingly important as children progress through school. Unlike informal learning experiences that often emerge naturally through play, homework imposes external constraints such as time pressure, performance expectations, and rigid structures, often placing parents in a position where they feel responsible for ensuring children complete assignments correctly and on time (e.g., [Silinskas et al., 2015b](#); [Wu et al., 2022](#)).

According to Grolnick et al. (e.g., [Grolnick, 2015](#); [Grolnick et al., 1997](#); [Lerner et al., 2022](#)), when parents experience pressure—either externally (e.g., from teachers or school policies) or internally (e.g., from their own anxieties about their child's success)—they are less likely to use constructive

motivational practices. Instead, such pressure can undermine their own sense of autonomy and competence, leading them to adopt more controlling practices to quickly alleviate the immediate stressor, rather than focusing on long-term developmental goals, such as fostering autonomy and competence in children (e.g., [Deci & Ryan, 1985](#); [Grolnick et al., 2002](#)). Beyond increasing control, pressure may also heighten parents' negative affect, particularly in evaluative contexts like homework, where children's performance may feel like a reflection of their own competence as a parent (e.g., [Pomerantz et al., 2005](#)), which could elicit heightened stress and frustration (e.g., [DiStefano et al., 2020](#)).

As a result, the structured and evaluative nature of formal activities, such as homework, may lead parents to become more controlling, prioritizing correct answers over exploration and independent problem-solving, while also displaying more negative (vs. positive) affect (e.g., [DiStefano et al., 2020](#); [Else-Quest et al., 2008](#); [Wu et al., 2022](#)). Experimental evidence supports this idea. [Grolnick et al., \(2002\)](#) found that when parents were placed in a high-pressure situation regarding children's performance, they were significantly more likely to exhibit controlling practices, limiting children's autonomy. Over time, this pressure-driven involvement can create a cycle of negative reinforcement, in which parents' heightened control diminishes children's sense of autonomy and competence, ultimately leading to reduced motivation and disengagement from math.

In contrast, it is possible that more informal and playful math activities allow parents to adopt less directive, more autonomy-supportive and affectively positive approaches. Using daily reports from parents of young elementary school children, [Wu et al., \(2022\)](#) compared parents' motivational practices across math homework and activities and found that parents' involvement in math activities (vs. homework) was characterized by less negative affect and more positive affect. Notably, parents' heightened negative affect in the homework context predicted children's reduced liking for math, lower preference for challenging math tasks, and lower math achievement over a year, even after controlling for their prior math adjustment.



4. Implications for educational recommendations and interventions

Although the dual-pathway model on the role of parenting in children's math learning requires additional empirical validation, it, along with the empirical evidence to date, highlights the importance of both cognitive

and motivational resources provided by parents. The model has important implications for educational recommendations and interventions aimed at enhancing children's math learning. Most notably, efforts to harness the power of parents should target both parents' cognitive and motivational practices. Although educating parents about these two types of practices and how to use them will be important, efforts should also address attributes of children, parents, and the learning context that can support parents in optimizing both the cognitive and motivational resources they confer on children. For example, math homework—as well as other formal math learning activities—should be designed with attention to the pressure that homework can induce in parents. This may involve giving parents choice in the types of problems on which they work with children or moving away from grading homework. While this approach may support parents' motivational practices, it does not necessarily support their cognitive practices. To this end, homework design should also include the attributes (e.g., structure and guidance) that can enhance parents' cognitive practices.



5. Conclusion

In conclusion, the dual-pathway model underscores the critical role of parents' provision of both cognitive and motivational resources in children's math learning. Parents' cognitive practices, such as advanced math talk and using prompts, directly contribute to the development of children's math skills. Simultaneously, motivational practices, including autonomy support and positive affect, foster children's math beliefs, motivation, and engagement. The importance of both the cognitive and motivational pathways highlights the need for educational interventions that address both cognitive and motivational aspects of parents' involvement. Future research should continue to explore the distinct and interactive effects of cognitive and motivational parenting practices in the math context, as well as the antecedents that shape these practices. Of particular importance is to understand the influence of the broader cultural, socioeconomic, and institutional contexts, which have been relatively under-examined in regard to parents' role in children's math learning but are likely of much significance. By understanding the multifaceted nature of parents' involvement in children's learning, educators can develop targeted strategies to support parents in fostering their children's math learning, ultimately contributing to improved motivation and achievement in math as children navigate a variety of learning environments over the school years and as adults into their careers.

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